

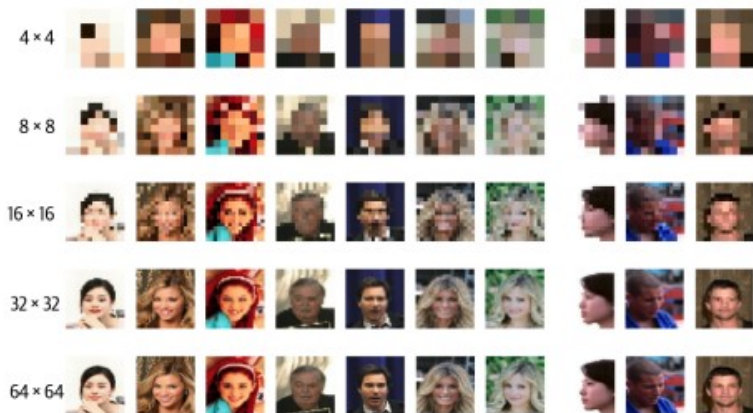


Figure: Images in the dataset can be compressed to lower resolution using interpolation

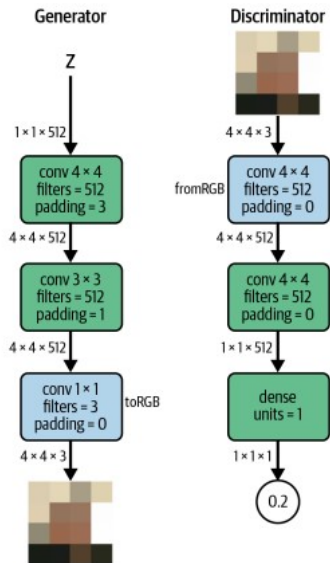


Figure: The generator and discriminator architectures for the first stage of the Pro-GAN training process

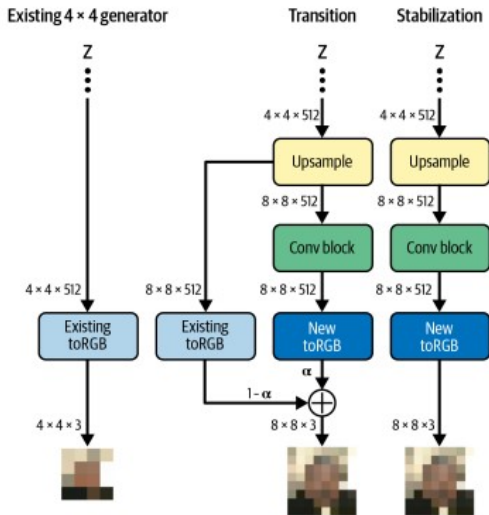


Figure: The ProGAN generator training process, expanding the network from 4×4 images to 8×8 (dotted lines represent the rest of the network, not shown)

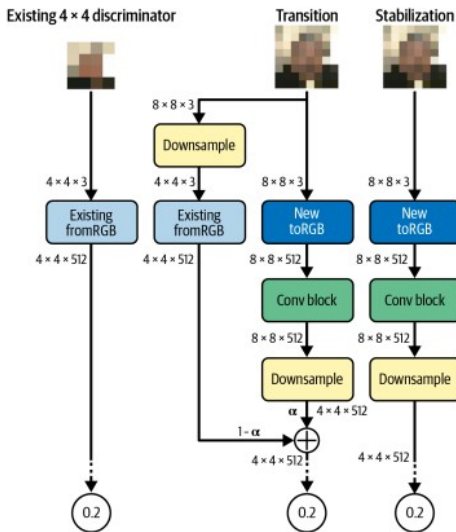


Figure: The ProGAN discriminator training process, expanding the network from 4×4 images to 8×8 (dotted lines represent the rest of the network, not shown)

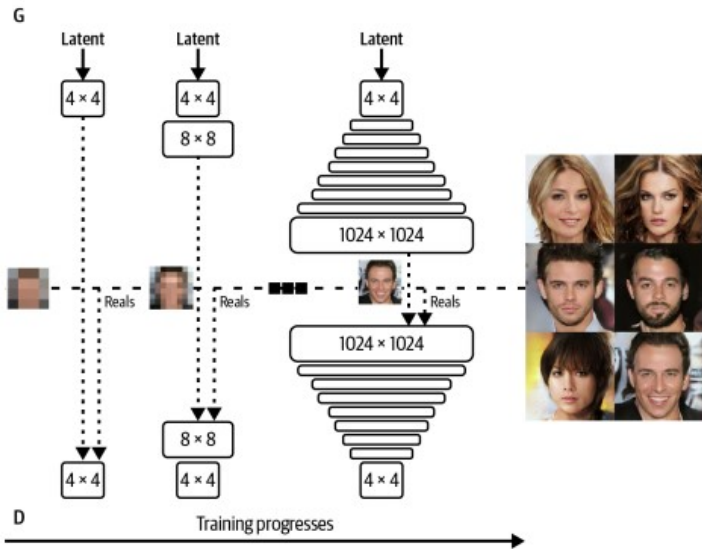


Figure: The ProGAN training mechanism, and some example generated faces

Generator	Act.	Output shape	Params
Latent vector	-	$512 \times 1 \times 1$	-
Conv 4×4	LReLU	$512 \times 4 \times 4$	4.2M
Conv 3×3	LReLU	$512 \times 4 \times 4$	2.4M
Upsample	-	$512 \times 8 \times 8$	-
Conv 3×3	LReLU	$512 \times 8 \times 8$	2.4M
Conv 3×3	LReLU	$512 \times 8 \times 8$	2.4M
Upsample	-	$512 \times 16 \times 16$	-
Conv 3×3	LReLU	$512 \times 16 \times 16$	2.4M
Conv 3×3	LReLU	$512 \times 16 \times 16$	2.4M
Upsample	-	$512 \times 32 \times 32$	-
Conv 3×3	LReLU	$512 \times 32 \times 32$	2.4M
Conv 3×3	LReLU	$512 \times 32 \times 32$	2.4M
Upsample	-	$512 \times 64 \times 64$	-
Conv 3×3	LReLU	$256 \times 64 \times 64$	1.2M
Conv 3×3	LReLU	$256 \times 64 \times 64$	590k
Upsample	-	$256 \times 128 \times 128$	-
Conv 3×3	LReLU	$128 \times 128 \times 128$	295k
Conv 3×3	LReLU	$128 \times 128 \times 128$	148k
Upsample	-	$128 \times 256 \times 256$	-
Conv 3×3	LReLU	$64 \times 256 \times 256$	74k
Conv 3×3	LReLU	$64 \times 256 \times 256$	37k
Upsample	-	$64 \times 512 \times 512$	-
Conv 3×3	LReLU	$32 \times 512 \times 512$	18k
Conv 3×3	LReLU	$32 \times 512 \times 512$	9.2k
Upsample	-	$32 \times 1024 \times 1024$	-
Conv 3×3	LReLU	$16 \times 1024 \times 1024$	4.6k
Conv 3×3	LReLU	$16 \times 1024 \times 1024$	2.3k
Conv 1×1	linear	$3 \times 1024 \times 1024$	51
Total trainable parameters			23.1M

Discriminator	Act.	Output shape	Params
Input image	-	$3 \times 1024 \times 1024$	-
Conv 1×1	LReLU	$16 \times 1024 \times 1024$	64
Conv 3×3	LReLU	$16 \times 1024 \times 1024$	2.3k
Conv 3×3	LReLU	$32 \times 1024 \times 1024$	4.6k
Downsample	-	$32 \times 512 \times 512$	-
Conv 3×3	LReLU	$32 \times 512 \times 512$	9.2k
Conv 3×3	LReLU	$64 \times 512 \times 512$	18k
Downsample	-	$64 \times 256 \times 256$	-
Conv 3×3	LReLU	$64 \times 256 \times 256$	37k
Conv 3×3	LReLU	$128 \times 256 \times 256$	74k
Downsample	-	$128 \times 128 \times 128$	-
Conv 3×3	LReLU	$128 \times 128 \times 128$	148k
Conv 3×3	LReLU	$256 \times 128 \times 128$	295k
Downsample	-	$256 \times 64 \times 64$	-
Conv 3×3	LReLU	$256 \times 64 \times 64$	590k
Conv 3×3	LReLU	$512 \times 64 \times 64$	1.2M
Downsample	-	$512 \times 32 \times 32$	-
Conv 3×3	LReLU	$512 \times 32 \times 32$	2.4M
Conv 3×3	LReLU	$512 \times 32 \times 32$	2.4M
Downsample	-	$512 \times 16 \times 16$	-
Conv 3×3	LReLU	$512 \times 16 \times 16$	2.4M
Conv 3×3	LReLU	$512 \times 16 \times 16$	2.4M
Downsample	-	$512 \times 8 \times 8$	-
Conv 3×3	LReLU	$512 \times 8 \times 8$	2.4M
Conv 3×3	LReLU	$512 \times 8 \times 8$	2.4M
Downsample	-	$512 \times 4 \times 4$	-
Minibatch stddev	-	$513 \times 4 \times 4$	-
Conv 3×3	LReLU	$512 \times 4 \times 4$	2.4M
Conv 4×4	LReLU	$512 \times 1 \times 1$	4.2M
Fully-connected	linear	$1 \times 1 \times 1$	513
Total trainable parameters			23.1M

Figure: The ProGAN generator and discriminator used to generate $1,024 \times 1,024$ – pixel CelebA faces



Figure: Generated examples from a ProGAN trained progressively on the LSUN dataset at 256×256 resolution

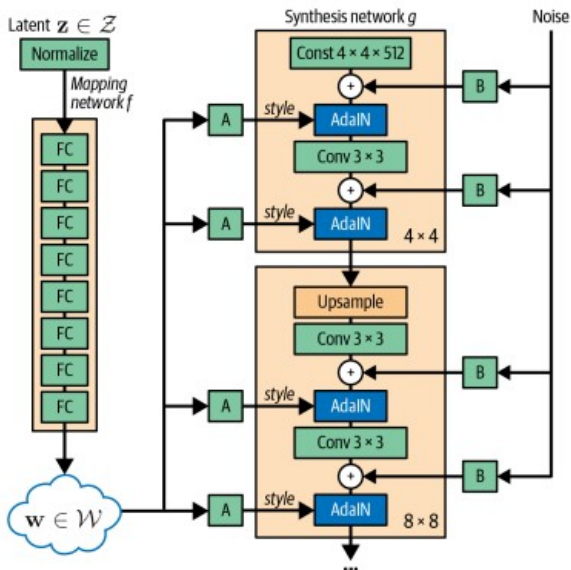


Figure: The StyleGAN generator architecture

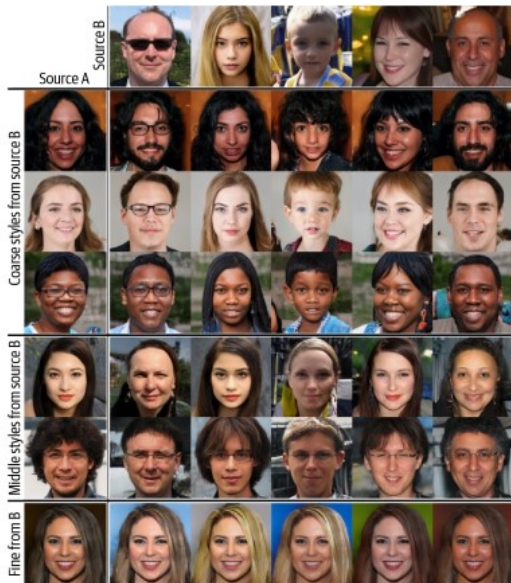


Figure: Merging styles between two generated images at different levels of detail



Figure: An artifact in a StyleGAN-generated image of a face

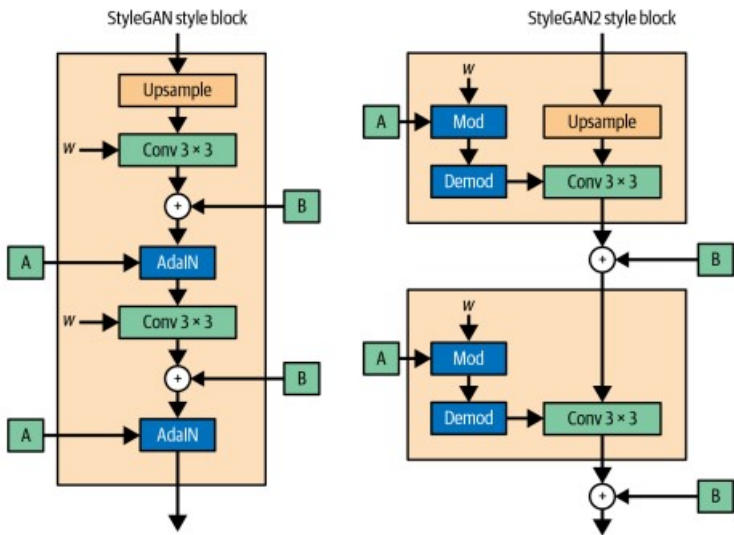


Figure: A comparison between the StyleGAN and StyleGAN2 style blocks

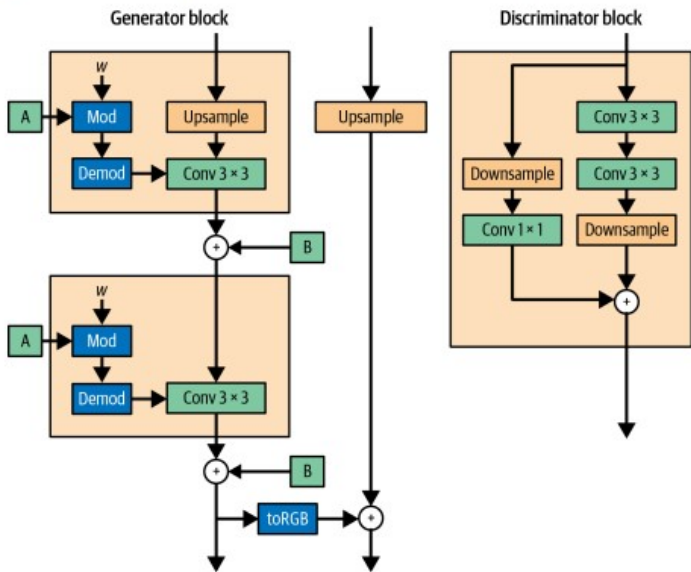


Figure: The generator and discriminator blocks in StyleGAN2

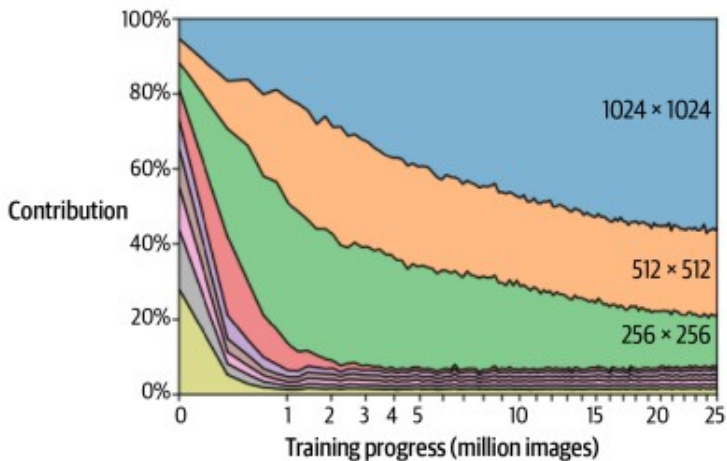


Figure: The contribution of each resolution layer to the output of the generator, by training time



Figure: Uncurated StyleGAN2 output for the FFHQ face dataset and LSUN car dataset

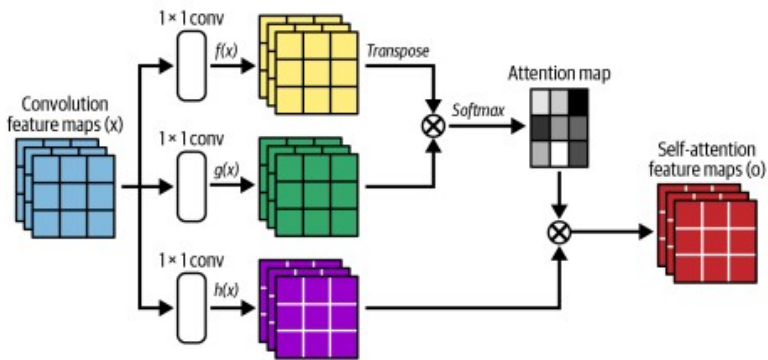


Figure: The self-attention mechanism within the SAGAN model



Figure: A SAGAN-generated image of a bird (leftmost cell) and the attention maps of the final attention-based generator layer for the pixels covered by the three colored dots (rightmost cells)



Figure: Examples of images generated by BigGAN



Figure: The truncation trick: from left to right, the threshold is set to 2, 1, 0.5, and 0.04

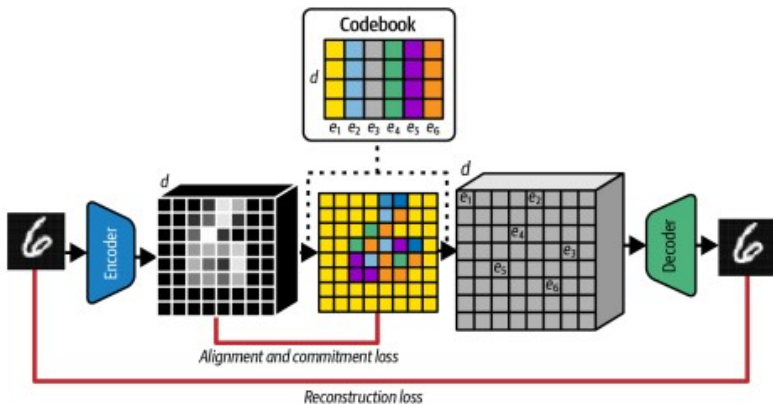


Figure: A diagram of a VQ-VAE

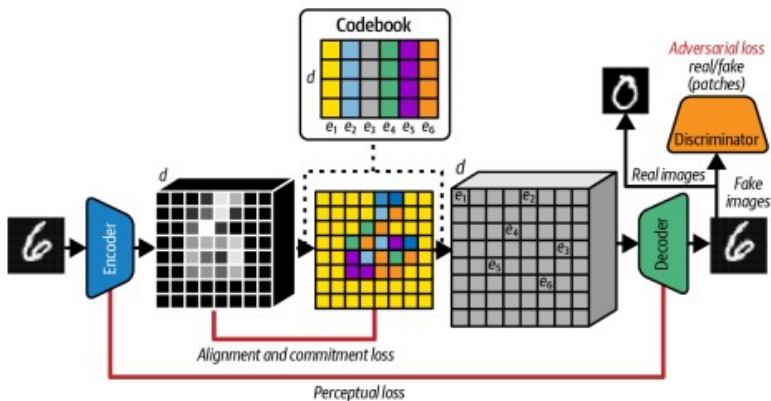


Figure: A diagram of a VQ-GAN: the GAN discriminator helps to encourage the VAE to generate less blurry images through an additional adversarial loss term

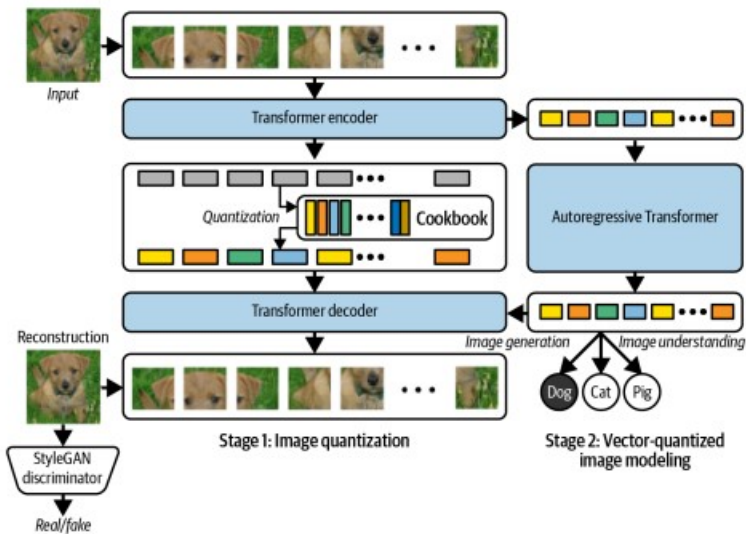


Figure: A diagram of a ViT VQ-GAN: the GAN discriminator helps to encourage the VAE to generate less blurry images through an additional adversarial loss term



Figure: Example images generated by a ViT VQ-GAN trained on ImageNet